Doing the Z80 sheet one step at a time

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This is the conversation about your posted KiCad files, written up so you can follow it without the tool chatter. Petr asked for it as a guide, not as a new design. The other note in this directory, `261103-what-went-wrong-and-how-to-do-it.adoc`, is the diagnosis of the files as you first posted them. Those files are kept in `as-posted/`.

What you posted

You gave an assistant one prompt that asked for a complete Z80 computer: schematic, board, routing, a monitor ROM, a bill of materials, and the write-up, all at once. The source drawing is Thomas Scherrer's simple Z80 system, the GIF at `http://www.z80.info/gfx/z80test.gif`.

The three files that came back look like a KiCad project. They are not a circuit.

- The schematic opens on a blank sheet. The parts are names only. There are no symbol drawings, no pins, and no wires, so the netlist is empty. A KiCad 6+ sheet has to carry the symbol drawings inside the file. Pointing at a library name is not enough.
- The board does not open. Four lines begin with `;`, which is not legal in that file. Taking those lines out still leaves a board with no pads.
- The project file itself is fine. It is not why the board fails.

The useful lesson is the method, not a cleverer prompt. One pile of jobs gives you three files that name a computer. Nobody opened them.

How to work with the assistant

Do it as a chat with an engineer, one checkable step at a time. After each step, open the file with KiCad, or with `kicad-cli` if you are on a machine that has it. Do not continue while the file you just wrote does not load.

A step is small enough to say what "done" means. For this sheet, done meant: it opens, the parts are real library symbols, ERC reports no errors, and a netlist shows the buses you expect. It did not mean a routed board.

If you use the desktop app rather than a terminal, the same rule holds. Save, open the schematic, run ERC from the Inspect menu, and look at the sheet before you ask for the next thing. When something fails, paste the message back and stop there. Do not ask the assistant to "just finish the PCB" on top of a sheet that has not been checked.

The assistant on this job was told the same thing, and it still ran ahead for a while. The pause in the middle was to say where the work actually was. That is worth doing on your side too.

What we did first

The posted files were copied to `as-posted/` before anything was overwritten. The diagnosis note describes those copies. The board in the project directory is still that posted board. It was not fixed and it was not routed.

Symbols and footprints came from the official KiCad 10.0.7 libraries. The old GitHub `kicad-symbols` repository is the KiCad 5 library and is archived. The current libraries are on GitLab, tagged `10.0.7`:

- https://gitlab.com/kicad/libraries/kicad-symbols
- https://gitlab.com/kicad/libraries/kicad-footprints

Only the parts this sheet uses were copied into `libs/`. The project files `sym-lib-table` and `fp-lib-table` point at those copies, so ERC can see them on a machine with no system libraries installed. The symbol drawings are also embedded in the schematic, which is what lets the sheet open at all.

This was done with KiCad 10.0.7. A current KiCad 10 should open it. A much older KiCad may not.

The parts, and where the library does not have your name

The GIF’s names were kept as values wherever the 10.0.7 library uses a different symbol. Nothing here was drawn by hand.

Ref	On the sheet	Why that symbol
U1	Value `Z84C00`, symbol `CPU:Z84C00xxP`	There is no plain `Z80` symbol. This is the CMOS part in a 40-pin DIP, pin-compatible with the NMOS Z80 on the GIF.
U2	Value `62256`, symbol `Memory_RAM:HM62256BLP`	The GIF's 32K RAM. Data pins on this symbol are named Q0–Q7. They are on the data bus.
U3	Value `27C256`, symbol `Memory_EPROM:27C256`	The GIF shows a 27C512. You had already asked for a 27C256. Pin 1 of a 27C256 is VPP, and it is tied to +5V. A15 does not go to the ROM. The map stays ROM at `0000–7FFF` and RAM at `8000–FFFF`.
U4	`82C55A`	The PIO. Ports A, B, and C also land on J2, J3, and J4, so those nets are real connector pins instead of a name hanging on one pin.
U5	Value `16C550`, symbol `Interface_UART:16550`	CS0 and CS1 are tied high. WR, RD, and /ADS are tied low, so only /WR and /RD strobe the chip. /BAUDOUT is wired to RCLK.
U6	`MAX232`	Charge-pump capacitors follow the MAX232 datasheet.
U7	Value `74HCT14`, symbol `74xx:74HC14`	The library has no 74HCT14. Same pinout.
U8	Value `74HCT32`, symbol `74xx:74LS32`	The library has no 74HC32 or 74HCT32. Same pinout. The value is the part you wanted; the drawing is the part the library has.
Y1, Y2	4 MHz and 1.8432 MHz oscillator symbols	Full-can oscillators, the four real pins only.

Ref	On the sheet	Why that symbol
J1	DE9 socket, wired as a DTE	Pin 2 RX, pin 3 TX, pin 4 DTR, pin 5 ground, pin 6 DSR. The footprint also has a shell pad named `SH`. The symbol has no pin for it, so the shell is not grounded. That is the one pad difference ERC does not complain about, and it is worth a decision when the board is done.

One KiCad 10 surprise, because it will bite a multi-unit part: a gate marked "not in the BOM" is dropped from the netlist entirely. U7 and U8 looked connected and were not, until every unit was left in the BOM. KiCad still lists each chip once.

The address decode matches the printed map, not a single chain of OR gates. RAM is selected when /MREQ is low and A15 is low. ROM is selected when /MREQ is low and A15 is high. The PIO answers an I/O cycle with A7 low (use ports 00–03). The UART answers an I/O cycle with A7 high (use 80–87). Reset is two Schmitt inverters. The 8255 RESET pin and the 16550 MR pin are active high, so they take the first inversion. The Z80 /RESET pin takes the second.

How to check the sheet

Open `Z80_Simple_Computer.kicad_pro`. The schematic is one A2 sheet. ERC on this machine, with KiCad 10.0.7, reports 0 errors and 0 warnings.

From a terminal, the same check is:

```
kicad-cli sch erc --severity-all Z80_Simple_Computer.kicad_sch
kicad-cli sch export netlist --format kicadsexpr -o /tmp/z80.net
Z80_Simple_Computer.kicad_sch
```

Things worth spotting on the netlist, or with the highlight tool in the schematic editor:

- U1 is present. The posted sheet had lost the processor.
- U3 pin 1 is on +5V, not on A15.
- A0–A14 and D0–D7 are common to the CPU, the RAM, and the ROM.
- /RAMCS is only the RAM chip select. /ROMCS is only the ROM chip enable.
- U5 pins WR, RD, and /ADS are on ground. /WR and /RD are on the Z80 strobes.

The buses are mostly net labels rather than long wires across the sheet. That is normal for this kind of drawing, and it is what the netlist is checking. The 100 nF capacitors are one per IC, drawn in a row along the bottom, not tucked beside each chip. The sheet says so.

The sheet can be regenerated with `python3 tools/build_schematic.py`. If you move a part in the editor and then run that script, the script wins. Either edit in KiCad and stop using the script, or change the script. Do not do both.

What this is not

The board file still does not open. There is no routing, no monitor ROM, and no bill of materials. Those were the rest of the original prompt, and they are the wrong next prompt.

A sensible next step is one of these, and then a check:

1. Make the posted board open, or throw it away and start a new empty board from this schematic.
2. Update the board from the schematic and look at the footprints. Do not route yet.
3. Only then place the parts. Through-hole, the footprints already named on the sheet.

Stop after each of those and look.